MILITARY SPECIFICATION SHEET

TUBING, PLASTIC, FLEXIBLE, CONVOLUTED, FLUORINATED ETHYLENE PROPYLENE, STANDARD CONVOLUTION

This specification has been approved by the Naval Air Systems Command, Department of the Navy

The complete requirements for procuring the tubing described herein shall consist of this document and the issue in effect of MIL-T-81914 (AS).

REQUIREMENTS:

Convolution type: Helical (see 3.3)

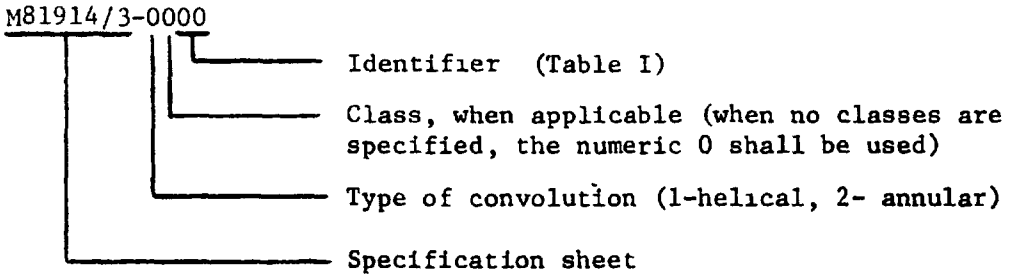
Construction details: Construction details, including available sizes, dimensions and tolerances, are located in Table I. Lengths shall be as specified by the procuring activity.

Continuous operating temperature: -67°C (-88°F) to +200°C (392°F).

Color: Unless otherwise specified, the supplied color shall be clear, natural.

Physical properties: General physical requirement values along with associated test conditions are located in Table II.

Part number: Consists of the basic number of this specification sheet and a dash number as shown below:



Standard convolution, Fluorinated Ethylene Propylene, standard wall 0.500 inch is identified as M81914 /3-1006.

☆U S GOVERNMENT PRINTING OFFICE 1981—703-023/6680 FSC 9330

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:22Z
Check the source to verify that this is the current version before use.

TABLE I

CONSTRUCTION DETAILS

Identifier	Max. Inside Dia.	Min. Inside Dia.	Max. Outer Dia.	Wall Thickness Max.	Convolutions Per Inch ± 1/2	Weight (Lbs) Per 100 feet Max.	Min. Bend Radius
**01	.187	.181	.320	.018	8	1.5	.500
**02	.281	.273	.414	.018	8	1.7	.750
**03	.312	.306	.450	.018	8	1.9	.750
**04	.375	.364	.510	.018	8	2.2	.875
**05	.437	.427	.571	.018	8	3.1	.875
**06	.500	.485	.650	.023	7	4.0	1.250
**07	.625	.608	.770	.023	7	4.8	1.500
**08	.750	.730	.930	.023	6	6.1	1.750
**09	.875	.860	1.073	.023	5	7.0	2.00
**10	1.000	.975	1.226	.023	5	8.5	2.37
**11	1.125	1.105	1.390	.023	5	9.3	2.37
**12	1.250	1.210	1.539	.023	4	10.9	2.75
**13	1.500	1.437	1.832	.023	4	12.6	3.38
**14	1.750	1.688	2.082	.023	4	14.8	3.88
**15	2.000	1.937	2.332	.023	4	16.8	4.25
**16	2.250	2.188	2.582	.023	4	17.6	5.38
**17	2.500	2.437	2.832	.023	4	20.0	6.00
**18	3.000	2.937	3.332	.023	4	25.1	7.00
**19	4.000	3.937	4.332	.023	4	38.1	9.50
**20	5.000	4.870	5.375	.023	4	43.7	11.0

** - The asterisks shall be replaced with convolution type and class designation.

Note: Unless otherwise specified all dimensions are in inches.

TABLE II

PHYSICAL PROPERTIES

PROPERTY	REQUIREMENT	TEST METHOD
Construction details	In accordance with Table I	4.6.1
Stress in psi @ 10% strain	225 to 700	4.6.2
Specific gravity, max.	2.18	4.6.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:22Z
Check the source to verify that this is the current version before use.

TABLE II (Continued)

PROPERTY	REQUIREMENT	TEST METHOD
Crush resistance, pounds Horizontal, min.	10	4.6.4
Low temperature flexibility	No cracking	4.6.5; 10,000 cycles
Heat shock	No dripping, flowing or cracking	4.6.6; 175 ±2°C (347 ±4°F) for 4 hours
Longitudinal change, %, max.	15	
Heat aging		4.6.7; 175 ±2°C (347 ±4°F) for 96 hours
Stress in psi at 10% strain, % of initial value	80	
Fluid resistance		4.6.8
Stress in psi at 10% strain, max. % of original value	80	
Flammability	Non-flammable 1/	4.6.9
Fungus resistance	Non-nutrient 1/	4.6.10
Dielectric breakdown	No breakdown, 12,000 volts	4.6.11

1/ See 6.4, Properties not tested.

Standard procurement: Procured as coils. Unless otherwise specified individual lengths shall be 18 feet.

Preparing Activity
Navy - AS
(Project No. 9330-N548-3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:22Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-T-81914/3(AS) TUBING, PLASTIC, FLEXIBLE, CONVOLUTED, FLUORINATED ETHYLENE PROPYLENE, STANDARD CONVOLUTION

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO (If "Yes", in what way?)

4 REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:22Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Engineering Center
Philadelphia, PA 19112

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE NAVY
DOD 316

U.S.MAIL

Engineering Specifications and Standards Department (Code X)
Naval Air Engineering Center
Philadelphia, PA 19112

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:22Z
Check the source to verify that this is the current version before use.